# Exercises week 1 – Answers

*8CM10 - Introduction to Quantum Physics and Structure of Matter*

## Special Relativity

1. The distance from New York to Los Angeles is about 5000 km and should take about 50 h in a car driving at 100 km/h.
   a. How much shorter than 5000 km is the distance according to the car travelers?

$$u = 100\frac{\text{km}}{\text{h}} = 28\ \text{m/s} \ll c$$

$$\sqrt{1 - u^2/c^2} \cong 1 - \frac{1}{2}\frac{u^2}{c^2} = 1 - \frac{1}{2}\frac{\left(28\frac{\text{m}}{\text{s}}\right)^2}{\left(3\cdot 10^8\frac{\text{m}}{\text{s}}\right)^2} = 1 - 4.3\cdot 10^{-15}$$

$$L = L_0\sqrt{1 - \frac{u^2}{c^2}} = L_0(1 - 4.3\cdot 10^{-15})$$

$$L_0 - L = 2.1\cdot 10^{-8}\ \text{m}$$

This is less than the wavelength of light

   b. How much less than 50 h do they age during the trip?

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}} \cong \Delta t_0(1 + 4.3\cdot 10^{-15})$$

$$\Delta t - \Delta t_0 = 4.3\cdot 10^{-15}\Delta t_0 = (4.3\cdot 10^{-15})(50\ \text{h})\left(3600\frac{\text{s}}{\text{h}}\right) = 7.7\cdot 10^{-10}\text{s}$$

2. How fast must an object move before its length appears to be contracted to one-half its proper length?

With $L = \frac{1}{2}L_0$, the length contraction formula gives $\frac{1}{2}L_0 = L_0\sqrt{1 - u^2/c^2}$, so

$$u = \sqrt{\frac{3}{4}}c = 2.6\cdot 10^8\ \text{m/s}$$

3. For what velocity $v$ is the total energy of a (relativistic) particle with mass $m$ equal to twice its rest energy?

The velocity $v$ where $E = 2mc^2$ is the solution of $\frac{mc^2}{\sqrt{1-v^2/c^2}} = 2mc^2$, which means $\sqrt{1 - v^2/c^2} = \frac{1}{2}$. That implies $v^2/c^2 = \frac{3}{4}$, which finally gives $v = \frac{\sqrt{3}}{2}c \approx 0.87c$.

4. According to observer $O$, a blue flash occurs at $x_b = 10.4\ m$ when $t_b = 0.124\ \mu s$, and a red flash occurs at $x_r = 23.6\ m$ when $t_r = 0.138\ \mu s$. According to observer $O'$, who is in motion relative to $O$ at velocity $u$, the two flashes appear to be simultaneous. Find the velocity $u$.

From the Lorentz transformations we know: $\Delta t' = \frac{\Delta t - ux/c^2}{\sqrt{1-u^2/c^2}}$. For $O'$ we find $\Delta t' = 0$, it must be true that $\Delta t - ux/c^2 = 0$. Thus:

$$u = \frac{c^2 \Delta t}{\Delta x} = \frac{c\left(300 \frac{\text{m}}{\mu\text{s}}\right)(0.138\ \mu\text{s} - 0.124\ \mu\text{s})}{(23.6\ \text{m} - 10.4\ \text{m})} = 0.32c$$

5. Two space exploration rockets approach the earth from opposing directions (say from $x$ and $-x$). From the frame of earth, each of the rockets has a speed of $0.4c$. In the frame of rocket 1, its length is $30\ m$. What length does the rocket have in the frame of rocket 2?

We need to use length contraction here:

$$L = L_0\sqrt{1 - u^2/c^2}$$

However, we need to be aware of what to use for $u$. In this case $u$ is the relative velocity of rocket 1 in the frame of rocket 2, but what is given is the velocities of the rockets in the frame of the earth. Therefore, we need to apply a velocity transformation:

$$v_{1\to R''} = \frac{v_{1\to E} - v_{2\to E}}{1 - v_{1\to E} \cdot v_{2\to E}/c^2}$$

Where $v_{1\to R''}$ is the velocity of rocket 1 in the frame of rocket 2, $v_{1\to E}$ is the velocity of rocket 1 in the frame of the earth, and $v_{2\to E}$ the velocity of rocket 2 in the frame of the earth.

In the frame of the Earth:

- Rocket 1 moves with a speed of $+0.4c$
- Rocket 2 moves with a speed of $-0.4c$

This gives:

$$v_{1\to R''} = \frac{0.4c - (-0.4c)}{1 - (0.4c)\cdot(-0.4c)/c^2} = \frac{0.8c}{1.16} \approx 0.6897c$$

Now, we can using the velocity for the length contraction:

$$L = L_0\sqrt{1 - v_{1\to R''}^2/c^2} = 30 \cdot \sqrt{1 - 0.6897^2} \approx 21.7 \text{ m}$$

## Relativistic Dynamics

6. The work-energy theorem relates the change in kinetic energy of a particle to the work done on it by an external force: $\Delta K = W = \int F dx$. Writing Newton's second law as $F = dp/dt$, show that $W = \int v\, dp$ and integrate by parts using the relativistic momentum to obtain:

$$K = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$$

$$W = \int F\, dx = \int \frac{dp}{dt} dx = \int dp \frac{dx}{dt} = \int v\, dp$$

Now apply integration by parts and integrate over all velocities:

$$\int_0^v v\, dp = pv - \int_0^v p\, dv = \frac{mv^2}{\sqrt{1 - \frac{v^2}{c^2}}} - \int_0^v \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}} dv$$

Substitute: $y \equiv 1 - \frac{v^2}{c^2}$ so that $dy = -\frac{2v}{c^2} dv$ and $dv = -\frac{c^2}{2v} dy$,
and $y(0) = 1,\ y(v) = 1 - \frac{v^2}{c^2}$

$$\int_0^v \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}} dv = \int_1^{1-\frac{v^2}{c^2}} -\frac{mc^2}{2\sqrt{y}} dy = -\left[mc^2\sqrt{y}\right]_1^{1-\frac{v^2}{c^2}} = -\left[mc^2\sqrt{1 - \frac{v^2}{c^2}} - mc^2\right]$$

Bringing this all together:

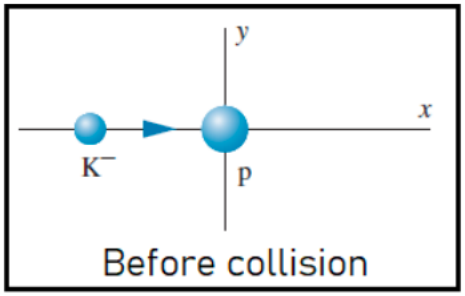
$$K = \int_0^v v\, dp = pv - \int_0^v p\, dv = \frac{mv^2}{\sqrt{1 - \frac{v^2}{c^2}}} - \int_0^v \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}} dv = \frac{mv^2}{\sqrt{1 - \frac{v^2}{c^2}}} + mc^2\sqrt{1 - \frac{v^2}{c^2}} - mc^2$$

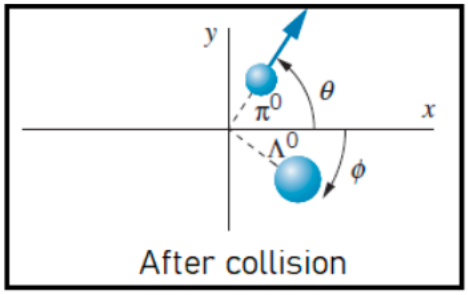
Multiply the second term with $\frac{\sqrt{1-\frac{v^2}{c^2}}}{\sqrt{1-\frac{v^2}{c^2}}}$:

$$= \frac{mv^2}{\sqrt{1-\frac{v^2}{c^2}}} + mc^2\sqrt{1-\frac{v^2}{c^2}}\cdot\frac{\sqrt{1-\frac{v^2}{c^2}}}{\sqrt{1-\frac{v^2}{c^2}}} - mc^2 = \frac{mv^2}{\sqrt{1-\frac{v^2}{c^2}}} + \frac{mc^2\left(1-\frac{v^2}{c^2}\right)}{\sqrt{1-\frac{v^2}{c^2}}} - mc^2$$

$$= \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}} - mc^2$$

7. Consider the reaction $K^- + p \rightarrow \Lambda^0 + \pi^0$ pictured below:




The rest masses and total energies are tabulated below:

| **Particle** | **Rest Masses** $(MeV/c^2)$ | **Total energy** $(MeV)$ |
|---|---|---|
| $K^-$ | 493.7 | 646.1 |
| $p$ | 983.3 | 983.3 |
| $\Lambda_0$ | 1115.7 | 1194.6 |
| $\pi_0$ | 135.0 | 389.8 |

Calculate the final momenta of $\pi_0$ and $\Lambda_0$, including their angles with respect to the $xy$-plane ($\theta$ and $\phi$ as defined in the picture).

We compute the momenta of all particles by using the energy-momentum relation:

$$p_i = \frac{1}{c}\sqrt{E_i^2 - (m_i c^2)^2}$$

Where $i$ denotes the particle. The results for all particles are:
$p_K = 416.8\ \text{MeV/c},\ p_p = 0\ \text{MeV/c},\ p_\Lambda = 426.9\ \text{MeV/c},\ p_\pi = 365.7\ \text{MeV/c}.$

At this point we have already computed the momenta, but we also want the angles. Using conservation of momentum before and after the collision we have two equations for the two unknown angles:

$p_K - p_\pi \cos\theta = p_\Lambda \cos\phi$ and $p_\pi \sin\theta = p_\Lambda \sin\phi$

Squaring both identities and adding yields:

$$p_\Lambda^2(\cos^2\phi + \sin^2\phi) = p_K^2 + p_\pi^2(\cos^2\theta + \sin^2\theta) - 2p_K p_\pi \cos\theta$$

$$p_\Lambda^2 = p_K^2 + p_\pi^2 - 2p_K p_\pi \cos\theta$$

Solving for $\theta$:

$$\theta = \arccos\left(\frac{p_K^2 + p_\pi^2 - p_\Lambda^2}{2p_K p_\pi}\right) = 65.7°$$

Solving for $\phi$:

$$\phi = \arcsin\left(\frac{p_\pi \sin\theta}{p_\Lambda}\right) = 51.3°$$

8. An electron is moving at a speed of $0.81c$. By how much must its kinetic energy increase to raise its speed to $0.91c$?

$$K_f - K_i = (E_f - mc^2) - (E_i - mc^2) = E_f - E_i$$

$$= \frac{mc^2}{\sqrt{1 - v_f^2/c^2}} - \frac{mc^2}{\sqrt{1 - \frac{v_i^2}{c^2}}} = \frac{0.511\ \text{MeV}}{\sqrt{1 - 0.91^2}} - \frac{0.511\ \text{MeV}}{\sqrt{1 - 0.81^2}} = 0.361\ \text{MeV}$$

9. What is the change in mass when $1g$ of copper is heated from 0 to $100°C$? The specific heat capacity of copper is $0.40\ J/g \cdot K$.

$$\Delta E = mc\Delta T = (1\text{g})(0.40\ \text{J/g·K})(100\ \text{K}) = 40\ \text{J}$$

$$\Delta m = \frac{\Delta E}{c^2} = \frac{40\ \text{J}}{9 \times 10^{16}\ \text{m}^2/\text{s}^2} = 4.4 \times 10^{-16}\ \text{kg}$$

## Photoelectric Effect

10. The work function $\phi$ (=energy to release one electron) for the photoelectric effect in potassium is 2.25 eV. For the case when light with a wavelength of 360 nm falls on the potassium, calculate
    a. The stopping potential $V_s$

The maximal kinetic energy of released electrons is given by $K_{max} = hf - \phi$, where $\phi$ (the work function) is the energy needed to remove the least tight bound electron

from the material and f is the frequency of the incoming light. The stopping potential $V_s$ is related to the maximal kinetic energy by $eV_s = K_{max}$, where e is the magnitude of the electric charge of an electron. Hence $eV_s = hf - \phi$.
The energy of a photon with wavelength $\lambda$ is $hf = \frac{hc}{\lambda}$. Since $\phi$ is given in electron volts (eV), we perform the computation in eV. In units eV and nm (nanometer), the value of $hc = 1240$ eV nm. For $\lambda = 360$ nm this yields a photon energy of $hf = \frac{1240}{360} = 3.44$ eV. This yields $K_{max} = 3.44 - 2.25 = 1.19$ eV, which means that $V_s = 1.19\ V$.

b. The kinetic energy and the velocity of the fastest of the emitted electrons

The maximal kinetic energy of the electrons is $K_{max} = 1.19$ eV $= 1.19 \cdot (1.602 \cdot 10^{-19})$ J $= 1.91 \cdot 10^{-19}$J. Since $K_{max} = \frac{1}{2}mv^2$, where the electron mass $m_e = 9.109 \cdot 10^{31}$ kg, we find for the velocity of the electrons $v = 6.5 \cdot 10^5 \frac{\text{m}}{\text{s}}$.

11. A homogeneous monochromatic light beam, with wavelength 400 nm, falls perpendicularly on a material with a work function of $\phi = 2.0$ eV. The beam intensity is $3.0 \cdot 10^9$ W/m$^2$. For the case when light with a wavelength of 360 nm falls on the potassium.
    a. Assuming that each photon releases one electron, compute the number of electrons emitted per m$^2$ per second.

    Energy of one photon with $\lambda = 400$ nm is $\frac{hc}{\lambda} = \frac{1240}{400} = 3.1\ eV$. This is more than the stopping potential $\phi = 2.0\ eV$, hence electrons will be released by the photo-electric effect. The energy of a photon in Joule is $3.1 \cdot 1.602\ 10^{-19}$ J $= 4.97 \cdot 10^{-19}$ J. Since an intensity of $3.0 \cdot 10^{-9}\ W/m^2$ means $3.0 \cdot 10^{-9}$ Joule per m$^2$ per s, there arrive $(3.0 \cdot 10^{-9})/(4.97 \cdot 10^{-19}) = 6.04 \cdot 10^9$ photons per m$^2$ per s. This leads to also $6.04 \cdot 10^9$electrons per m$^2$ per s.

    b. Calculate the total absorbed energy

    The release of one electron requires an energy $\phi$, that is absorbed by the material. Hence the total absorbed energy per m$^2$ and per second is $6.04 \cdot 10^9 * 2.0$ eV $= 6.04 \cdot 10^9 * 2.0 * 1.602 \cdot 10^{-19}$Jm$^{-2}$s$^{-1} = 1.94 \cdot 10^{-9}$ Wm$^{-2}$
    Alternatively: From the energy of each photon (3.1 eV) is 2.0 eV used to release the electron. This energy stays in the material, i.e. is absorbed. Hence a fraction $\frac{2.0}{3.1}$ of the total incoming energy is absorbed. This yields: $\frac{2.0}{3.1} * 3.0 \cdot 10^{-9} = 1.94 \cdot 10^{-9}$ Wm$^{-2}$ for the total absorbed energy.

c. Calculate the kinetic energy of the photoelectrons

The maximal kinetic energy of the photons is $K_{max} = hf - \phi = 3.1 - 2.0\ \text{eV} = 1.1\ \text{eV} = 1.1 * 1.602 \cdot 10^{-19}\ \text{J} = 1.76 \cdot 10^{-19}\ \text{J}$

12. On a graph of $K_{max}$ versus $f$ for the photoelectric effect, what is the physical significance of

The formula is:

$$K_{max} = hf - \phi$$

a. The slope
   The slope is $h$, Planck's constant
b. The intercept with the horizontal axis $f$
   The intercept with the horizontal axis is the cutoff frequency $f_c$
c. The intercept with the vertical axis $K_{max}$
   The intercept with the vertical axis is at $-\phi$, where $\phi$ is the work function

## Thermal Radiation

13. What is the temperature of a black sphere with a diameter of 10 cm which is emitting a total of 100 W thermal radiation? How much mass is lost each year by being radiated away? (use $E = mc^2$ to relate the mass loss with the emitted energy)

The intensity of the radiation from the sphere equals 100W, divided by the area of the surface $A = 4\pi R^2 = 4\pi \cdot 0.05^2 = 0.0314\ \text{m}^2$. Hence, $I = \frac{100}{A} = 3.18 \cdot 10^3\ \text{W m}^{-2}$. Using Stefan-Boltzmann's law we now find for the temperature $T$ of the sphere

$$T = \sqrt[4]{\frac{I}{\sigma}} = \sqrt[4]{\frac{3.18 \cdot 10^3}{5.6703 \cdot 10^{-8}}} = 487\text{K}$$

We first compute the mass-loss due to 1 sec energy radiation. In 1 sec the sphere emits $\Delta E = 100\ \text{J}$, which leads to a mass loss $m = \Delta E/c^2 = \frac{100}{(2.998 \cdot 10^8)^2} = 1.1125\ 10^{15}\ \text{kg}$. The mass loss in one year is then $60 * 60 * 24 * 365 * 1.1125 \cdot 10^{15} = 3.5 \cdot 10^{-8}\ \text{kg}$.

14. Assuming that the sun is a spherical black body with a radius $r = 7 \cdot 10^8$ m. The intensity I of the solar radiation at the surface of the earth (which is $1.5 \cdot 10^{11}$ m from the sun) is $1.4 \cdot 10^3$ W/m². Calculate the temperature of the surface of the sun.

The intensity of the solar radiation at a certain distance $R_1$ to the sun is given by the total energy output of the sun, divided by $4\pi R_1^2$, the surface area of a sphere with radius $R_1$. This means that the intensity at distance $R_1$ scales with $R_1^{-2}$. Let $R_A = 1.5 \cdot 10^{11}$ m be the

distance of the earth to the sun and let $R = 7 \cdot 10^8$ m be the radius of the sun. The Intensity of the radiation at the surface of the sun is then given by:

$$I_{sun} = \frac{R_A^2}{R^2} I_E = \left(\frac{1.5 \cdot 10^{11}}{7 \cdot 10^8}\right)^2 * 1.4 \cdot 10^3 = 6.4 \cdot 10^7 \ \mathrm{Wm^{-2}}$$

Using Stefan Boltzmann's law $I = \sigma T^4$ we then obtain for the surface temperature of the sun:

$$T = \sqrt[4]{\frac{I_{sun}}{\sigma}} = \sqrt[4]{\frac{6.4 \cdot 10^7}{5.6703 \cdot 10^{-8}}} = 5800 \ \mathrm{K}$$

15. The radiation law of Rayleigh Jeans says that $I(\lambda) = \frac{2\pi c}{\lambda^4} kT$, while Planck's radiation law is $I(\lambda) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$. Show that for large wavelengths ($\lambda \to \infty$) the radiancy given by Planck's law approximates the Rayleigh Jeans radiancy.
*Hint: for small $x$ it holds that $e^x \approx 1 + x$*

If $\lambda \to \infty$ then $\frac{hc}{k\lambda T} \to 0$. Hence, using the hint with $x \equiv \frac{hc}{k\lambda T}$:

$$e^{\frac{hc}{k\lambda T}} - 1 \to \frac{hc}{k\lambda T} \qquad \text{for} \qquad \lambda \to \infty$$

For Planck's radiation law that means that for $\lambda \to \infty$:

$$R_{Planck}(\lambda) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{k\lambda T}} - 1} \to \frac{2\pi hc^2}{\lambda^5} \frac{k\lambda T}{hc} = \frac{2\pi ckT}{\lambda^4} = R_{R.J.}(\lambda)$$

16. The law of Stefan-Boltzmann for the total intensity *I* emitted by the surface of a black body with temperature T says: $I = \sigma T^4$. This law is consequence of Planck's radiation law: integrating the radiancy $R(\lambda)$ given by Planck's radiation law over all wavelengths $\lambda$ yields the total intensity $I$.
Derive the law of Stefan-Boltzmann and give the formula for the constant σ.
Compute the numerical value of σ.
*Hint: put in the integral $x = \frac{hc}{\lambda kT}$ and use that $\int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}$.*

Planck's radiation law says that:

$$R(\lambda) = \frac{c}{4} \frac{8\pi}{\lambda^4} \frac{hc}{\lambda} \frac{1}{e^{\frac{hc}{k\lambda T}} - 1} = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{k\lambda T}} - 1}$$

The total intensity $I$ is found by integrating $R(\lambda)$ over all possible wavelengths $\lambda$:

$$I = \int_0^\infty R(\lambda)d\lambda = \int_0^\infty \frac{2\pi hc^2}{\lambda^5}\frac{1}{e^{\frac{hc}{k\lambda T}}-1}d\lambda$$

To compute this integral, we introduce the new variable: $x = \frac{hc}{k\lambda T}$. Hence $\lambda = \frac{1}{x}\frac{hc}{kT}$, and consequently $d\lambda = \frac{-1}{x^2}\frac{hc}{kT}dx$. Substitution in the integral gives:

$$I = \int_\infty^0 2\pi hc^2 \frac{x^5k^5T^5}{h^5c^5}\frac{1}{e^x-1}\frac{-1}{x^2}\frac{hc}{kT}dx = 2\pi\frac{k^4T^4}{h^3c^2}\int_0^\infty \frac{x^3}{e^x-1}dx$$

By using the hint, we can solve this integral:

$$I = 2\pi\frac{k^4T^4}{h^3c^2}\frac{\pi^4}{15} = \frac{2\pi^5k^4}{15h^3c^2}T^4$$

This is Stefan-Boltzmann's law $I = \sigma T^4$, with

$$\sigma = \frac{2\pi^5k^4}{15h^3c^2} = 5.6703\cdot 10^{-8}\ \mathrm{Wm^{-2}K^{-4}}$$